

A large industrial steam turbine is shown in a factory setting. The turbine's complex structure, featuring numerous blades and a large central hub, is the central focus. A worker in a light blue shirt and dark pants stands to the right, providing a sense of scale to the massive machine. The scene is illuminated by warm, industrial lights.

STEAM TURBINES

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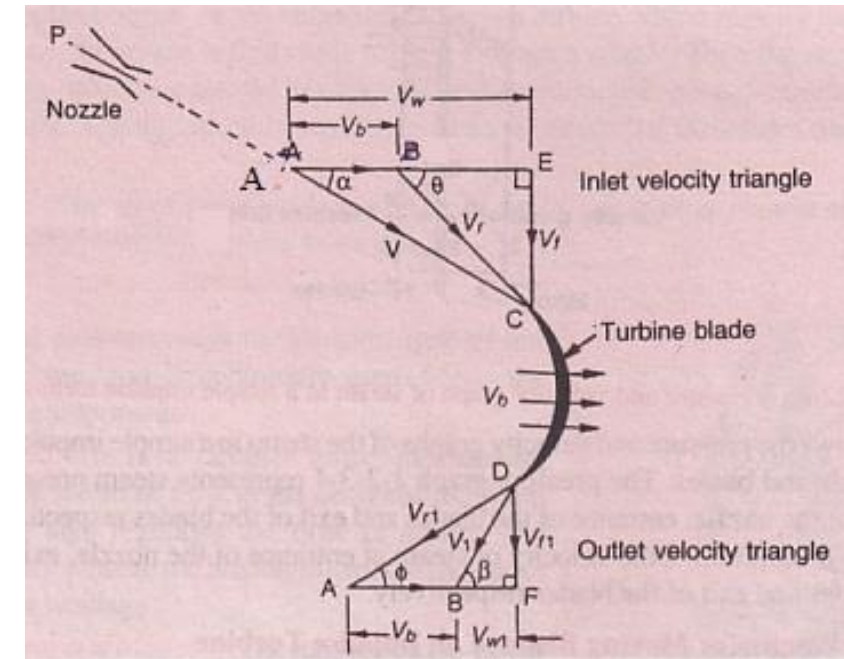
A steam turbine is a prime mover in which the potential energy of the steam is transformed into kinetic energy and later in its turn is transformed into the mechanical energy of rotation of the turbine shaft.

APPLICATIONS OF STEAM TURBINES

- Power Generation
- Transport

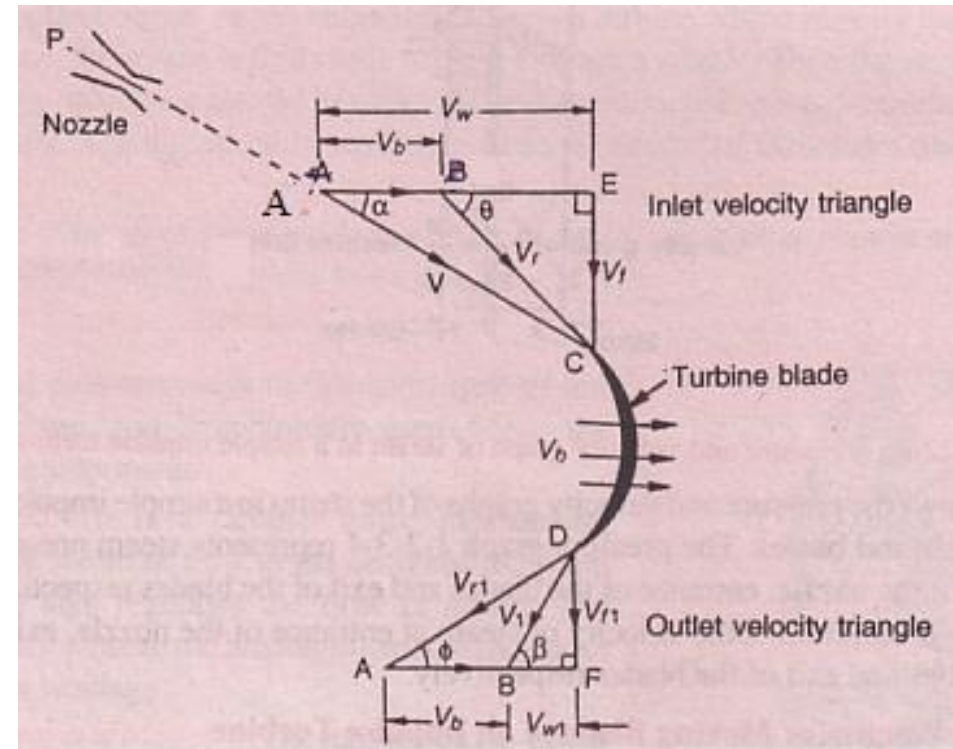
Velocity triangles for impulse turbine

- In an impulse turbine, the steam jet after leaving the nozzle impinges on one end of the blade.
- The jet then glides over the inside surface of the blade and finally leaves from the other edge
- The jet enters and leaves the blades tangentially for shockless entry and exit.

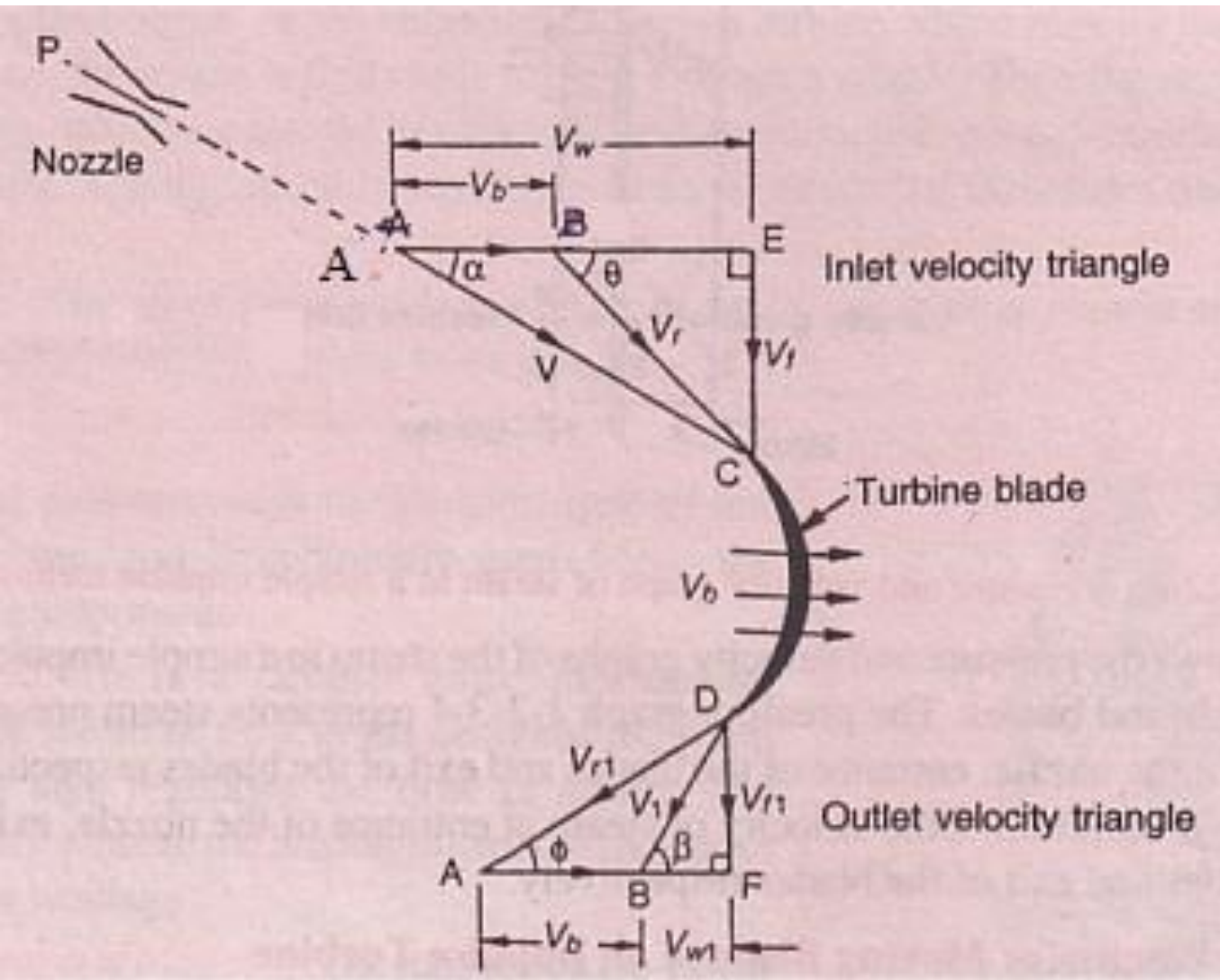


Velocity triangles for impulse turbine

- Consider a steam jet entering a curved blade after leaving the nozzle at **C**.
- Let the jet glides over the inside surface and leaves the blade at **D**
- Let us draw the velocity triangles at inlet and outlet tips of the moving blade



Velocity triangles for impulse turbine



V_b = Linear velocity of the moving blade (AB)

V = Absolute velocity of steam entering the moving blade (AC)

V_r = Relative velocity of jet to the moving blade (BC). It is the vectorial difference between V_b and V .

V_f = Velocity of flow at entrance of the moving blade. It is the vertical component of V .

V_w = Velocity of whirl at entrance of the moving blade. It is the horizontal component of V .

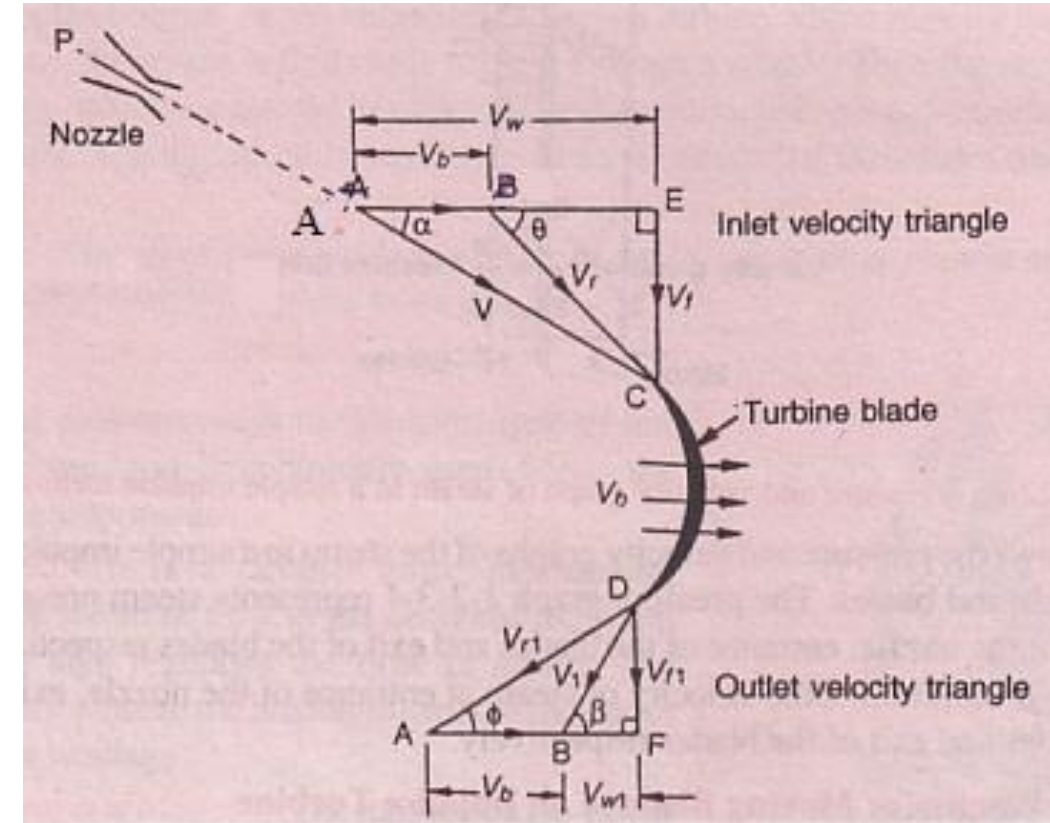
θ = Angle which the relative velocity of jet to the moving blade (V_r) makes with the direction of motion of the blade,

α = Angle with the direction of motion of the blade at which the jet enters the blade.

V_1 V_{r1} , V_{f1} , V_{w1} , β , ϕ = Corresponding values at exit of the moving blade.

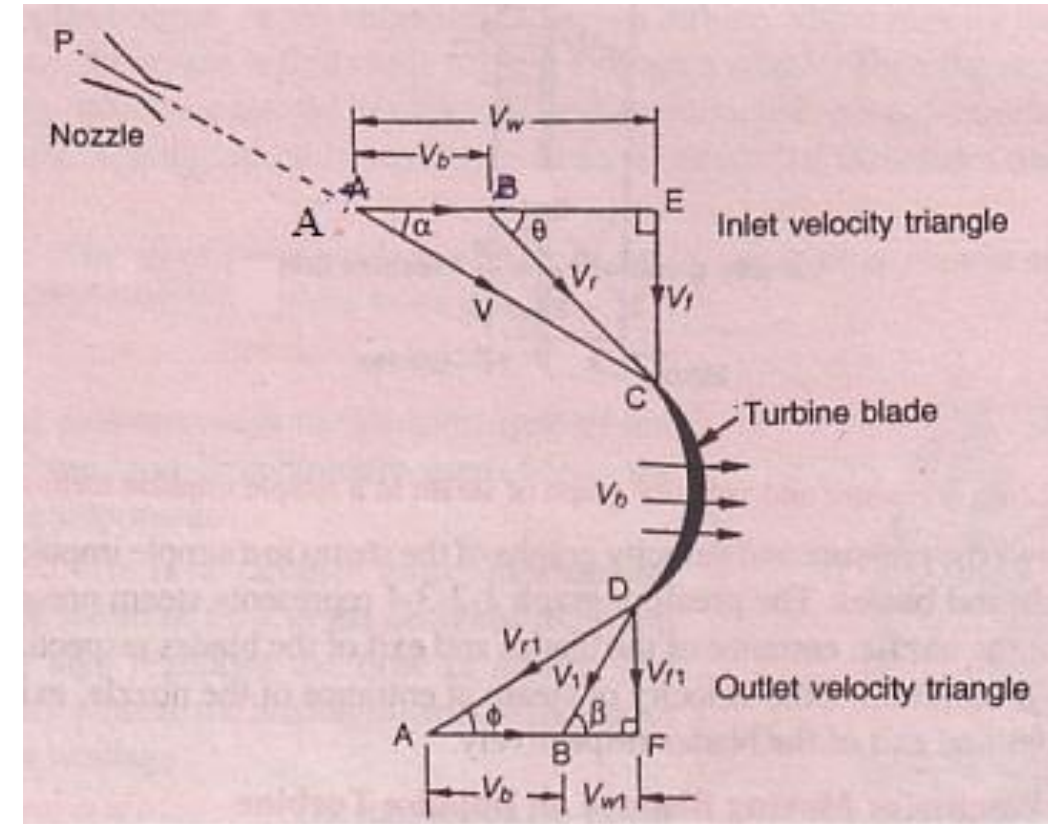
Velocity triangles for impulse turbine

- PC is the axis of the nozzle, which delivers the steam jet with a high velocity (V) at an angle α with the direction of motion of the blade.
- The jet impinges on a series of turbine blades mounted on the runner disc.
- The vertical component of V (i.e. EC) which does no work on the blade, is known as velocity of flow (V_f).
- It causes the steam to flow through the turbine and also an axial thrust on the rotor.



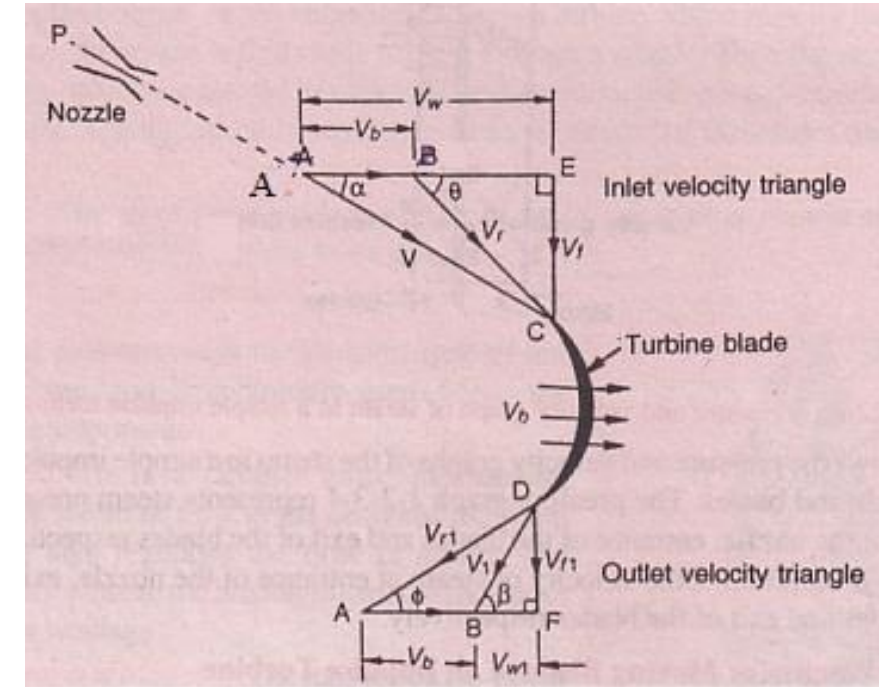
Velocity triangles for impulse turbine

- The horizontal component of V (represented by AE) is known as velocity of whirl at inlet (V_w).
- The linear velocity or mean velocity of the blade (i.e. V_b) is represented by AB in magnitude and direction.
- The length BC represents the relative velocity (V_r) of the steam jet with respect to the blade.



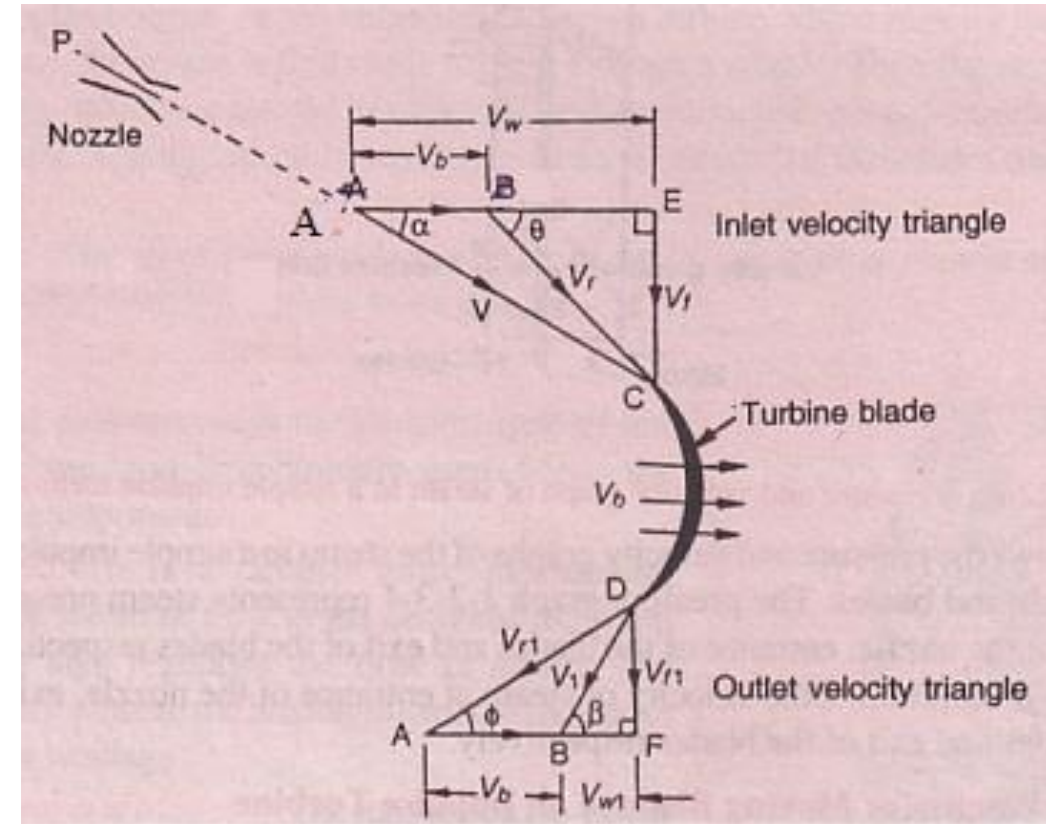
Velocity triangles for impulse turbine

- The jet glides over and leaves the blade with a relative velocity V_{r1} represented by DA.
- The absolute velocity of jet (V_1) as it leaves the blade, is represented by DB inclined at an angle β with the direction of the blade motion.
- The horizontal component of V_1 (represented by BF) is known as velocity of whirl at exit (V_{w1}).
- The vertical component of V_1 (represented by DF) is known as velocity of flow at exit (V_{f1}).



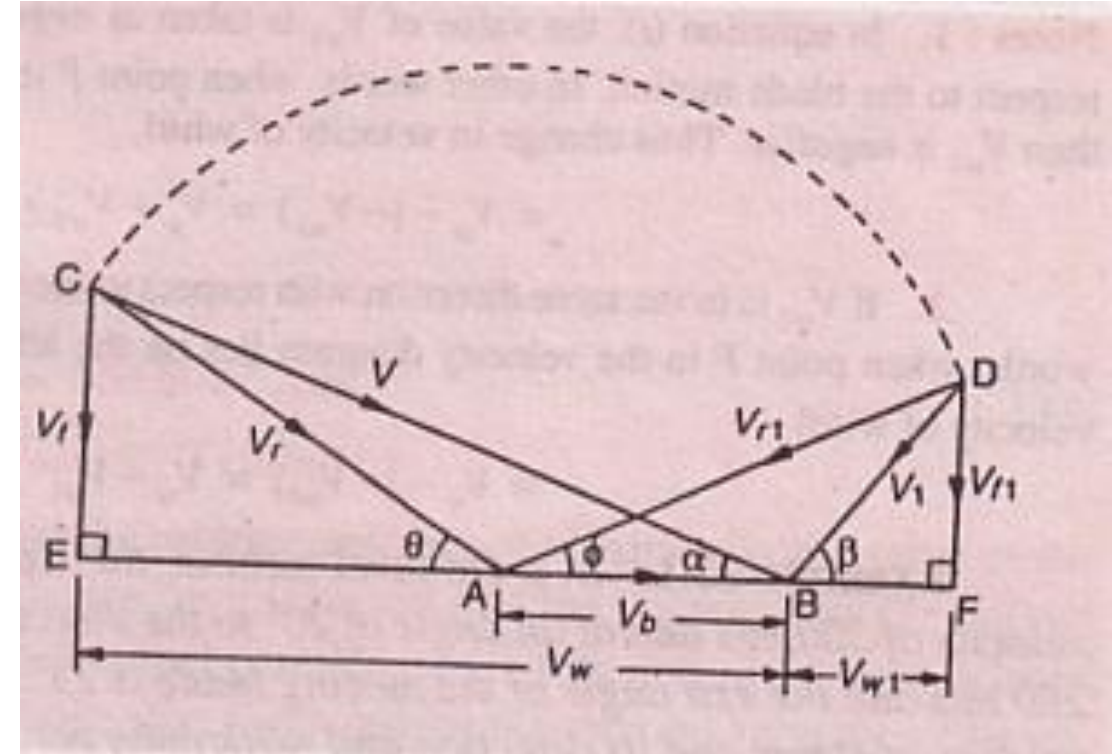
Velocity triangles for impulse turbine

- The inlet triangle of velocities is represented by AEC
- The outlet triangle by AFD
- The relations between inlet and outlet velocity triangle (until and unless given) is $V_r = V_{r1}$



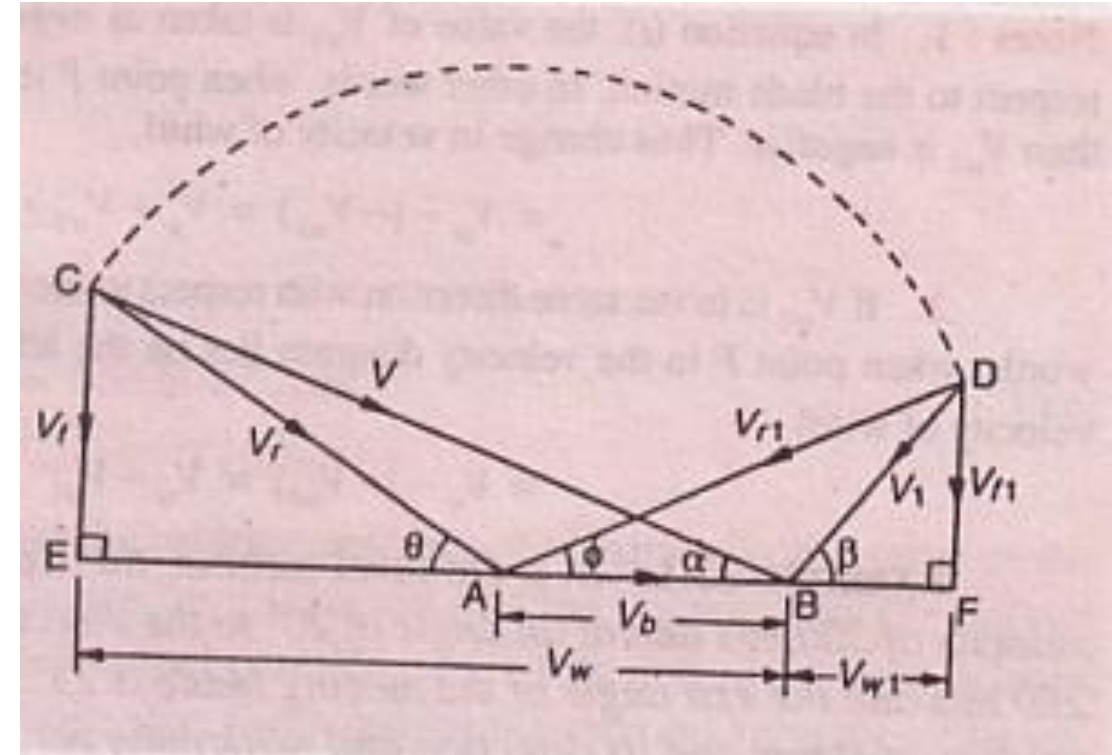
Combined Velocity Triangle for Moving Blades

- A combined velocity triangle for the moving blade can be drawn, for solving problems on steam turbines
- Draw a horizontal line, and cut off AB equal to velocity of blade (V_b) to some suitable scale.
- At B , draw a line BC at an angle α with AB .



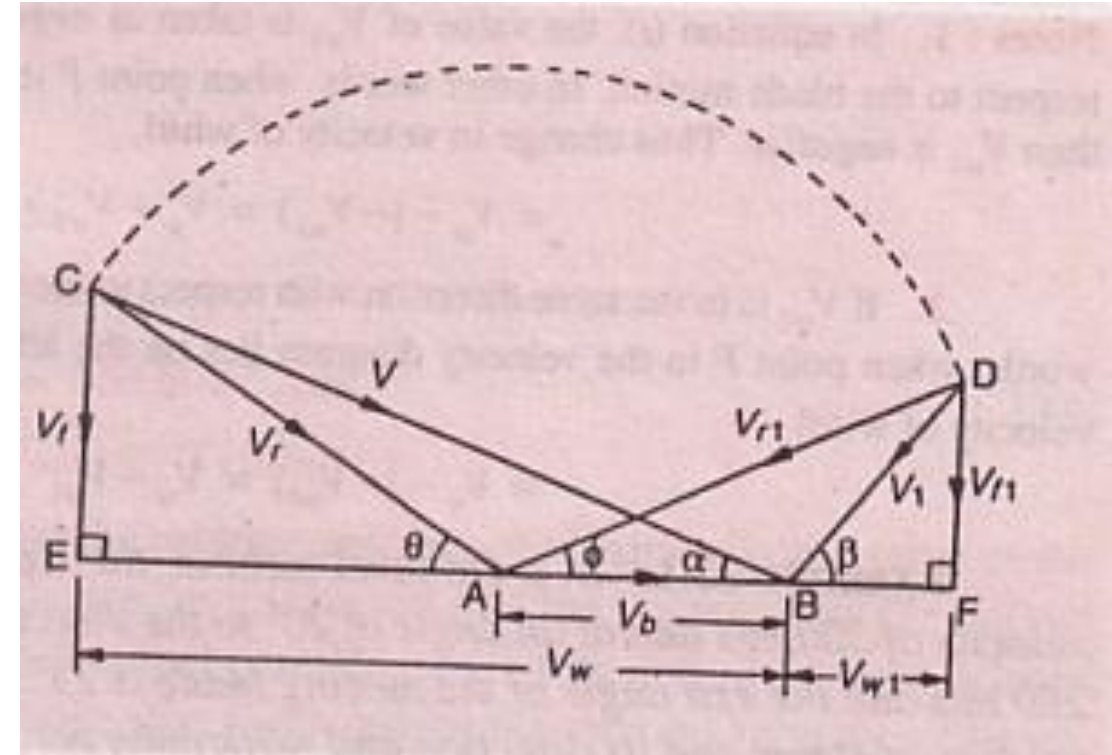
Combined Velocity Triangle for Moving Blades

- Cut off BC equal to V (i.e. velocity of steam jet at inlet of the blade) to the scale.
- Join AC, which represents the relative velocity at inlet (V_r).
- At A, draw a line AD at an angle ϕ with AB.
- With A as centre and radius equal to AC, draw an arc meeting the line through A at D, such that $AC = AD$ or $V_r = V_{r1}$



Combined Velocity Triangle for Moving Blades

- Join BD, which represents velocity of jet at exit (V_1) to the scale.
- From C and D, draw perpendiculars meeting the line AB produced at E and F respectively.
- EB and CE represents the velocity of whirl and velocity of flow at inlet (V_w and V_f) to the scale.
- BF and DF represents the velocity of whirl and velocity of flow at outlet (V_{w1} and V_{f1}) to the scale.



Force on Blades

- Consider an impulse turbine working under the action of a steam jet. Let us draw a combined velocity triangle

- Let m = Mass of the steam flowing through the turbine in kg/s

- $(V_w + V_{w1})$ = Change in the velocity of whirl in m/s.

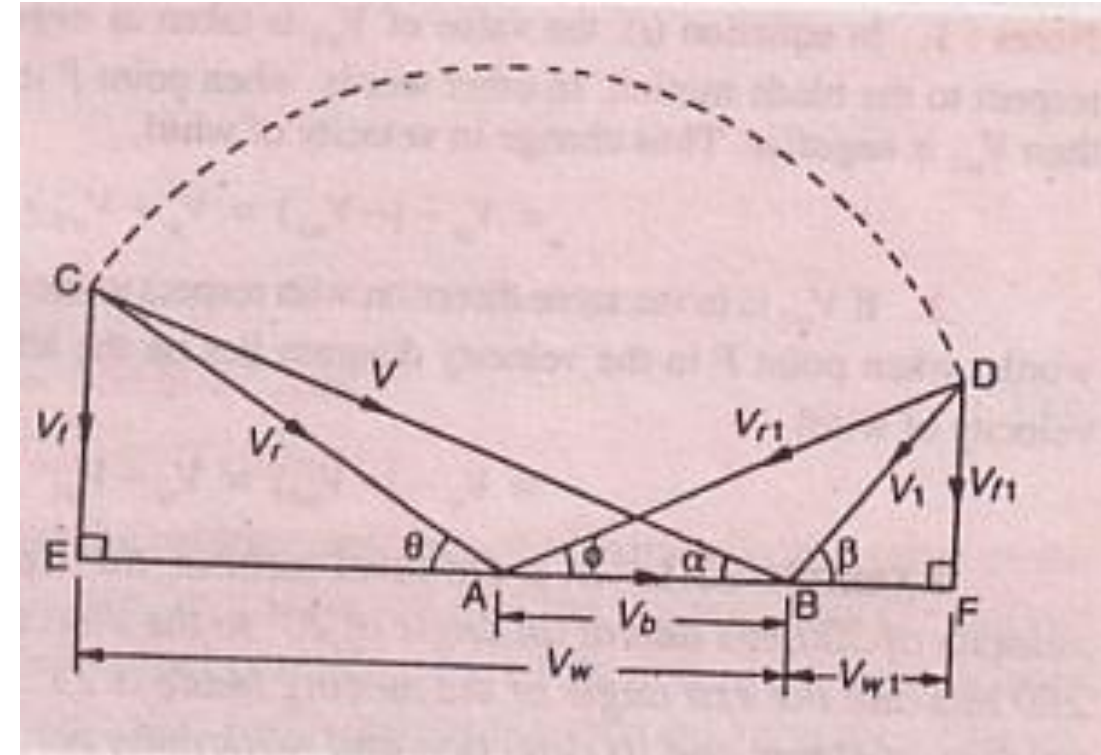
- Force in the direction of motion of the blades

F_x = Mass of steam flowing per second x Change in
 the velocity of whirl

$$= m[V_w - (-V_{w1})]$$

(the value of V_{w1} is taken as negative because of the opposite direction of V_{w1} with respect to the blade motion.)

$$= m[V_w + V_{w1}] = m \times EF \text{ N}$$



Work done by the blades

- Work done in the direction of motion of the blades

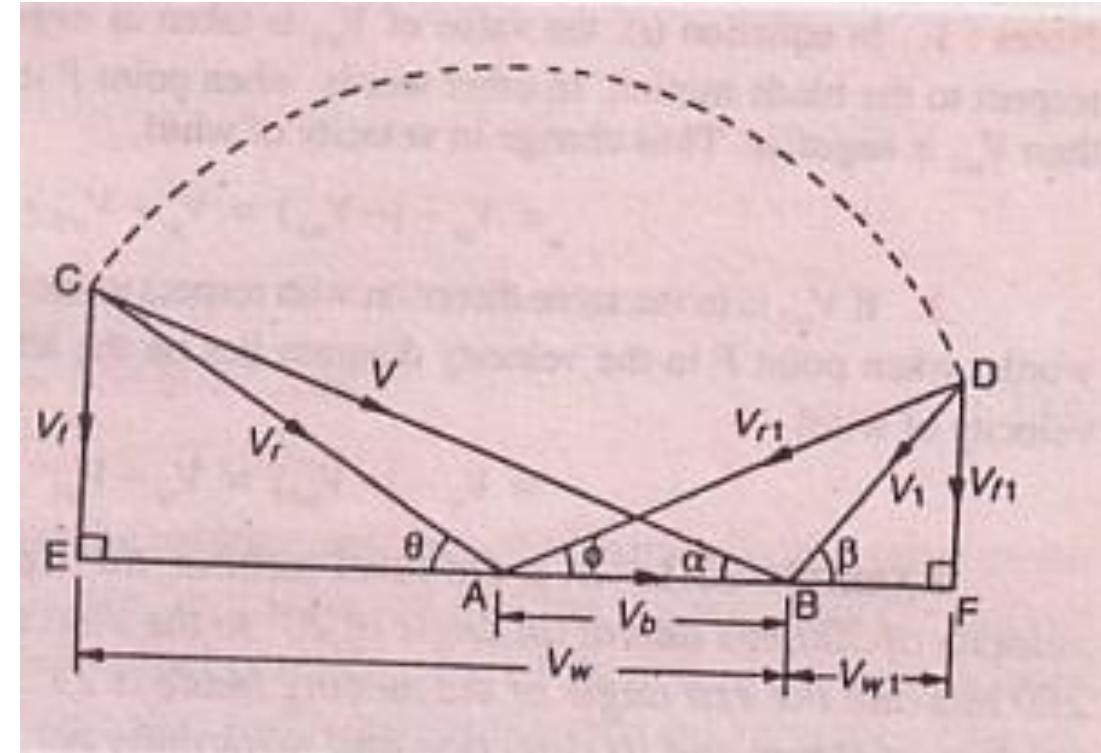
$$= \text{Force} \times \text{Distance}$$

$$= m[V_w + V_{w1}] V_b \quad \text{N-m/s}$$

$$= m \times EF \times AB \quad \text{N-m/s}$$

- Power produced by the turbine

$$P = m[V_w + V_{w1}] V_b \quad \text{W}$$



Axial thrust on the wheel

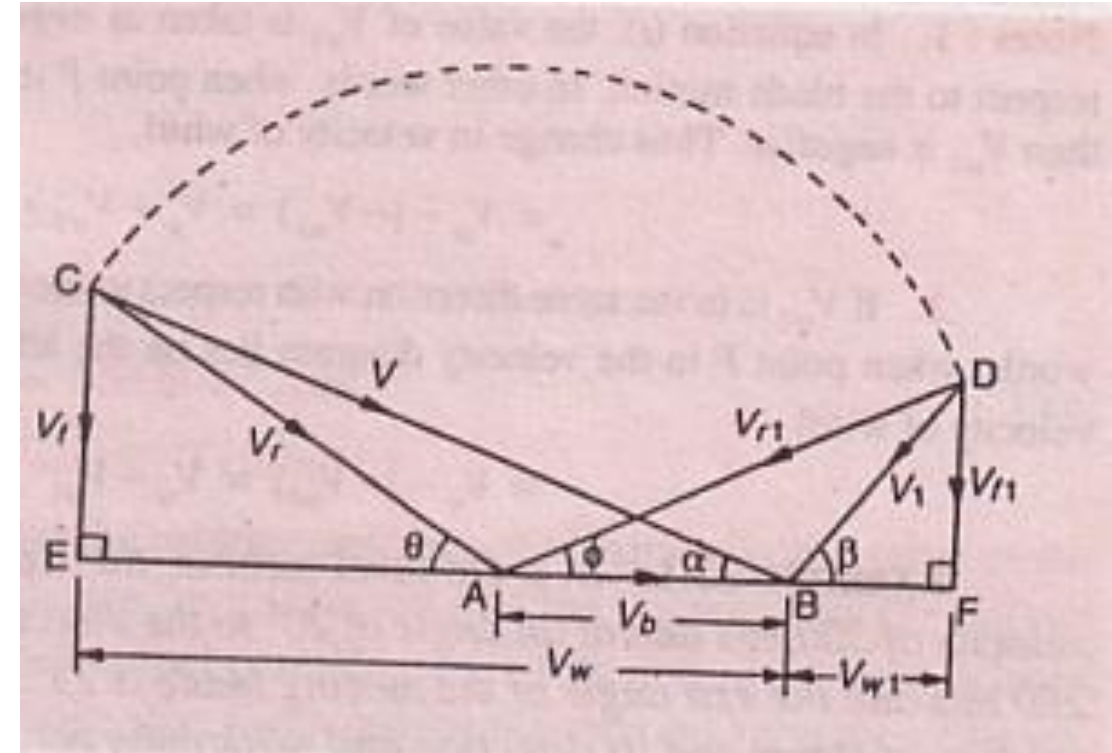
- The axial thrust on the wheel which is due to the difference of velocities of flow at inlet and outlet.

- Axial thrust on the wheel

$F_Y = \text{Mass of steam flowing per second}$
 $\times \text{Change in the velocity of}$
 flow

$$= m(V_f - V_{f1}) = m (CE - DF) \text{ N}$$

- When the blade friction is negligible, $V_f = V_{f1}$, axial thrust is zero



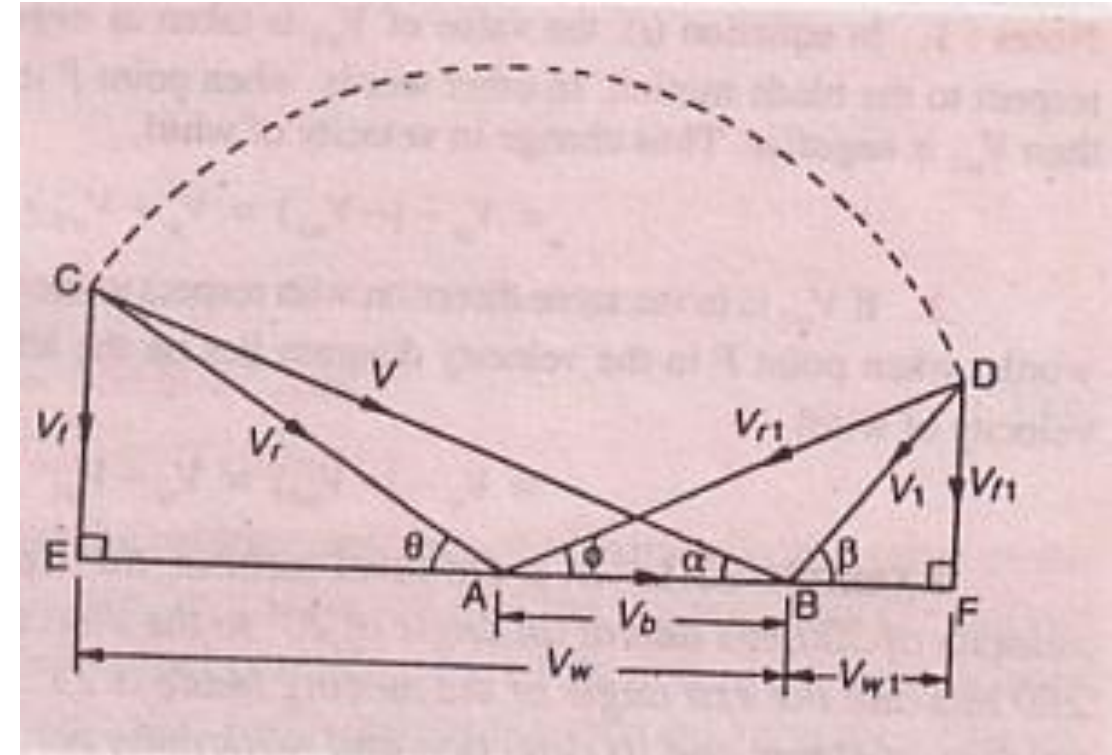
Blade or Diagram efficiency

Blade or Diagram efficiency

$$\eta_b = \frac{\text{Work done on the blades}}{\text{Energy supplied to the blades}}$$

$$\eta_b = \frac{m[V_w + V_{w1}] V b}{\frac{m V^2}{2}}$$

$$\eta_b = \frac{2 [V_w + V_{w1}] V b}{V^2}$$



Effect of friction on blades

- Energy is converted into heat due to friction.

$$Q_{friction} = m (Vr^2 - Vr1^2) \text{ W}$$

Blade velocity co-efficient (friction Factor) K

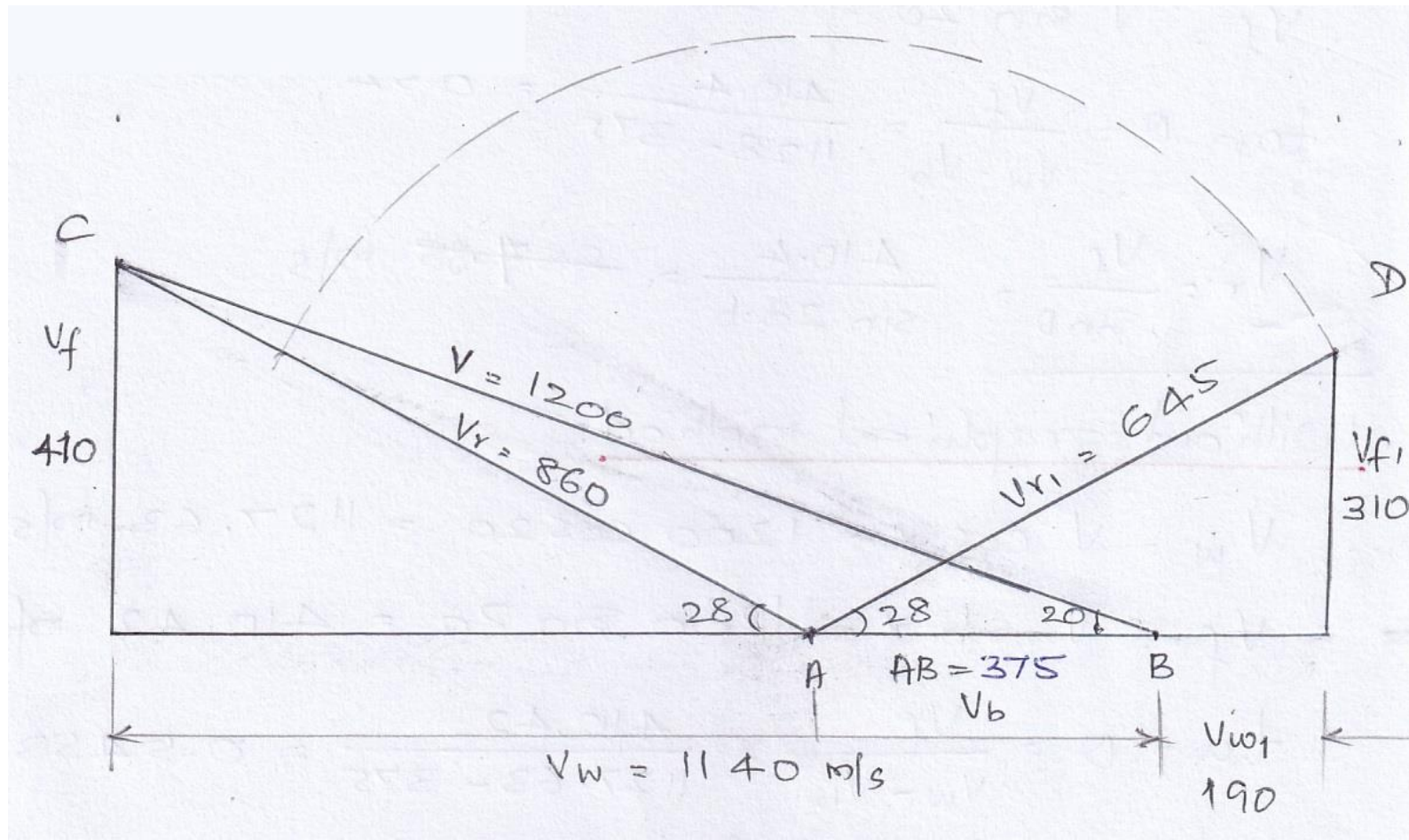
$$K = \frac{V_{r1}}{V_r}$$

An impulse turbine has exit velocity of steam from a converging nozzle equal to 1200 m/s . The nozzle makes an angle of 20° to the tangent of rotating wheel. The mean blade velocity is 375 m/s . The blades are symmetrical. The mass of steam flowing through the turbine is 0.5 kg/s . The blade velocity co-efficient is 0.75 . Find 1) Blade angles 2) Force on the wheel 3) Axial thrust 4) power developed.

Scale of the velocity diagram 1 cm = 100 m/s.

$$V = 1200 \text{ m/s} \quad \alpha = 20^\circ, \quad V_b = 375 \text{ m/s}, \quad K = \frac{V_{r1}}{V_r} = 0.75, \quad m = 0.5 \text{ kg/s}$$

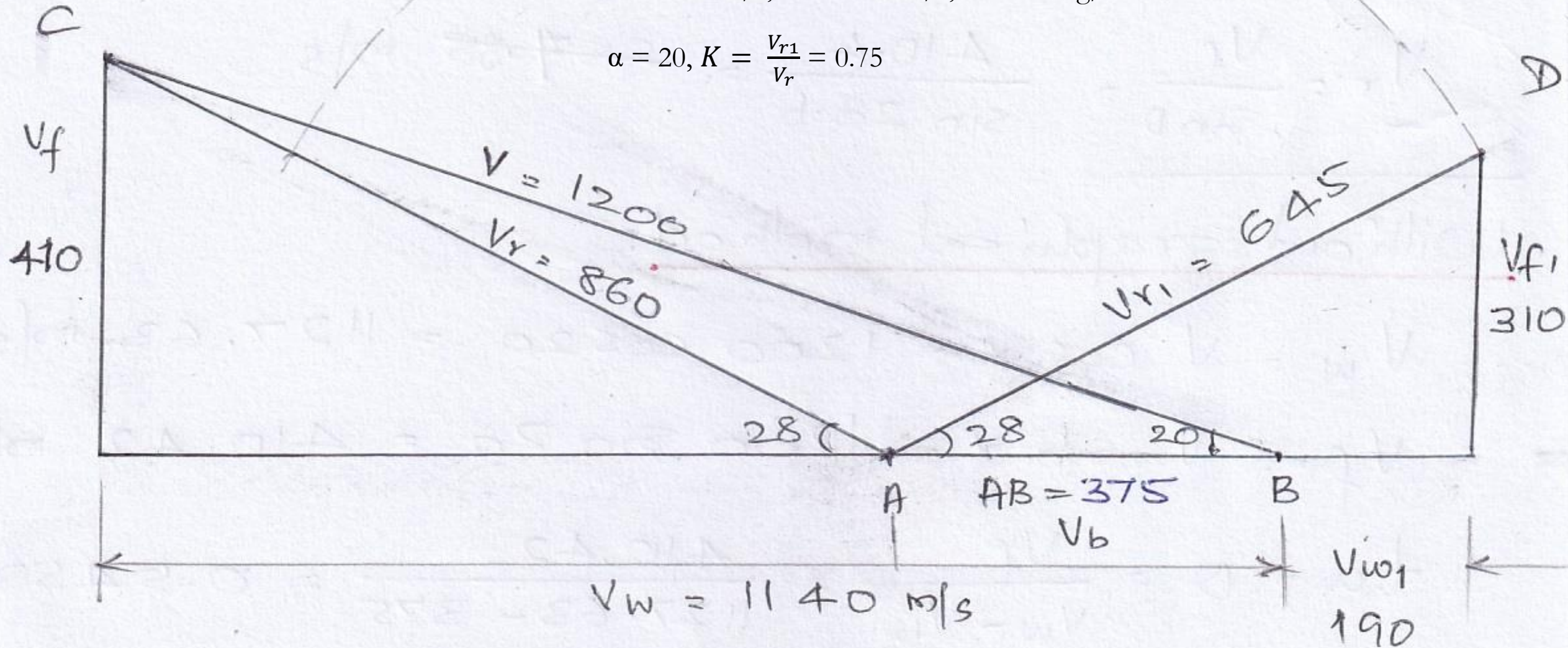
$\theta = \phi$ (symmetrical.)

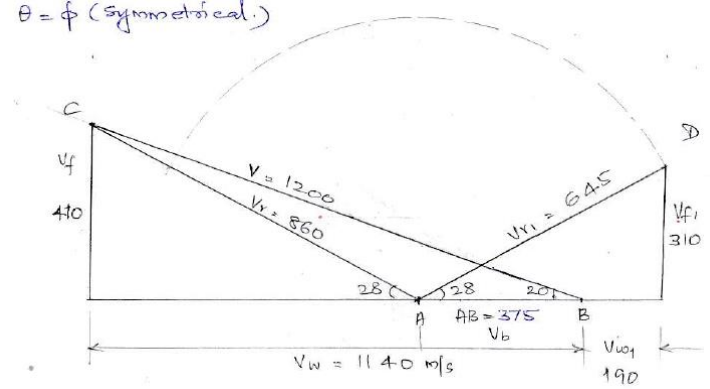


Scale of the velocity diagram 1 cm = 100 m/s

$V = 1200$ m/s, $V_b = 375$ m/s, $m = 0.5$ kg/s

$\alpha = 20^\circ$, $K = \frac{V_{r1}}{V_r} = 0.75$





$$\theta = \phi = 28^\circ, \quad V_w = 1140 \text{ m/s}, \quad V_{w1} = 190 \text{ m/s}$$

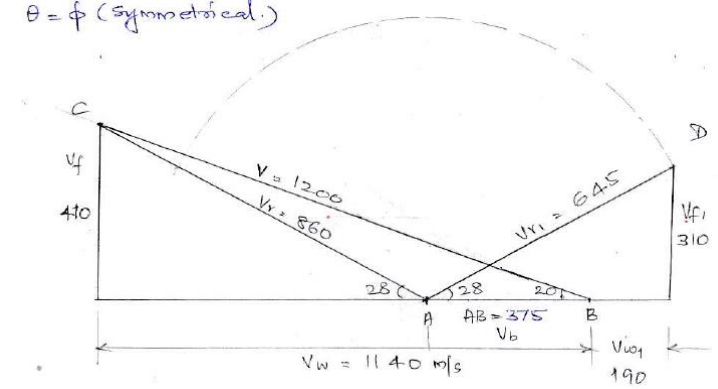
$$V_f = 410 \text{ m/s}, \quad V_{f1} = 310 \text{ m/s}.$$

1) Blade angle $\theta = \underline{28^\circ}$

2) Force on the wheel $F_x = m(V_w + V_{w1})$

$$= 0.5(1140 + 190)$$

$$= \underline{665 \text{ N}}$$



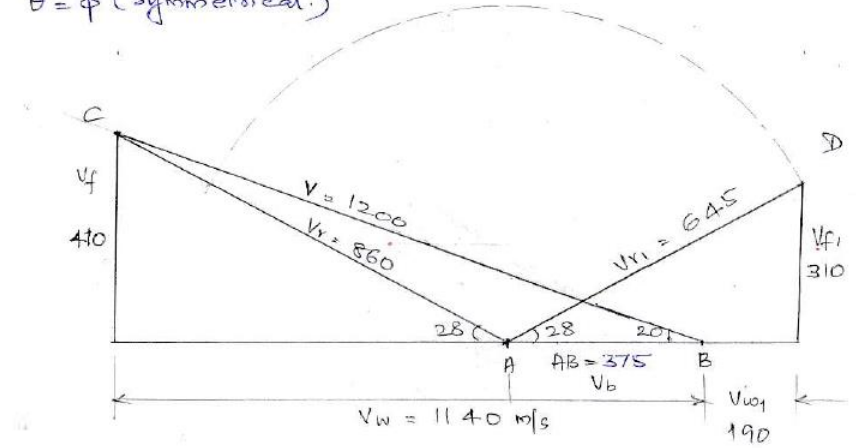
3) Axial Thrust on the wheel.

$$F_y = m (V_f - V_{f1}) = 0.5 (410 - 310) \\ = \underline{50 \text{ N}}$$

A) Power developed. $P = m (V_w + V_{w1}) V_b$

$$= 0.5 (1140 + 190) 375 = 249375 \text{ W}$$

$$= \underline{249.38 \text{ kW}}$$



Without graphical method.

$$V_w = V \cos \alpha = 1200 \cos 20 = 1127.63 \text{ m/s}$$

$$V_f = V \sin \alpha = 1200 \sin 20 = 410.42 \text{ m/s}$$

$$\tan \theta = \frac{V_f}{V_w - V_b} = \frac{410.42}{1127.63 - 375} = 0.5453$$

$$\theta = 28.60^\circ$$

$$V_r = \frac{V_f}{\sin \theta} = \frac{410.42}{\sin 28.6} = 857.38 \text{ m/s}$$

$$V_{r1} = 0.75 \times 857.38 = 643.03 \text{ m/s}$$

$$V_{f1} = V_{r1} \sin \theta = 643.03 \sin 28.6 = 307.82 \text{ m/s}$$

$$V_{w1} = (V_{r1} \cos 28.6) - 37 = (643.03 \cos 28.6) - 375$$

$$= 189.57 \text{ m/s}$$

An impulse turbine has exit velocity of steam from a converging nozzle equal to 900 m/s . The nozzle makes an angle of 20° to the tangent of rotating wheel. The mean blade speed is 400 m/s . The blades are equiangular (symmetrical). The mass of steam flowing thro' the turbine is 1200 kg/hr . Blade velocity coefficient $k = 0.85$
Find 1) blade angles, 2) axial thrust on blades
3) power developed.

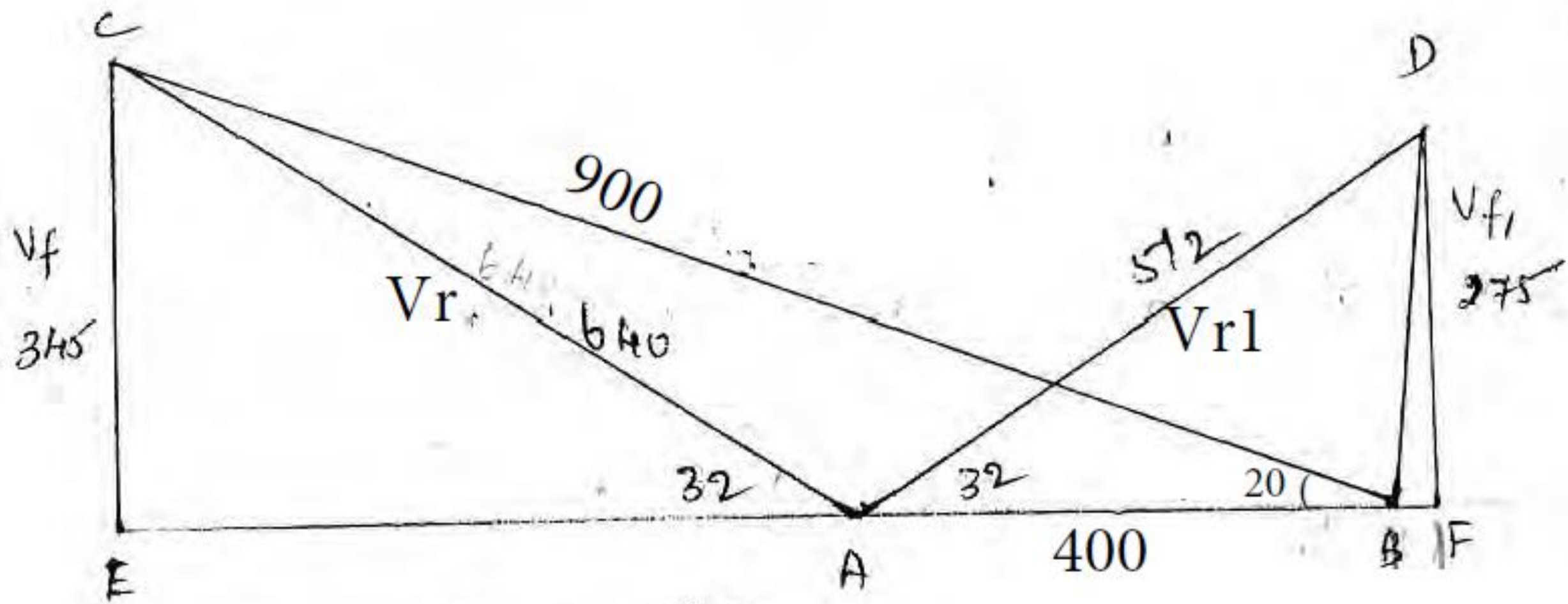
Scale of velocity diag; $1 \text{ cm} = 100 \text{ m/s}$.

$$V = 900 \text{ m/s} \quad \alpha = 20^\circ \quad V_b = 300 \text{ m/s}$$

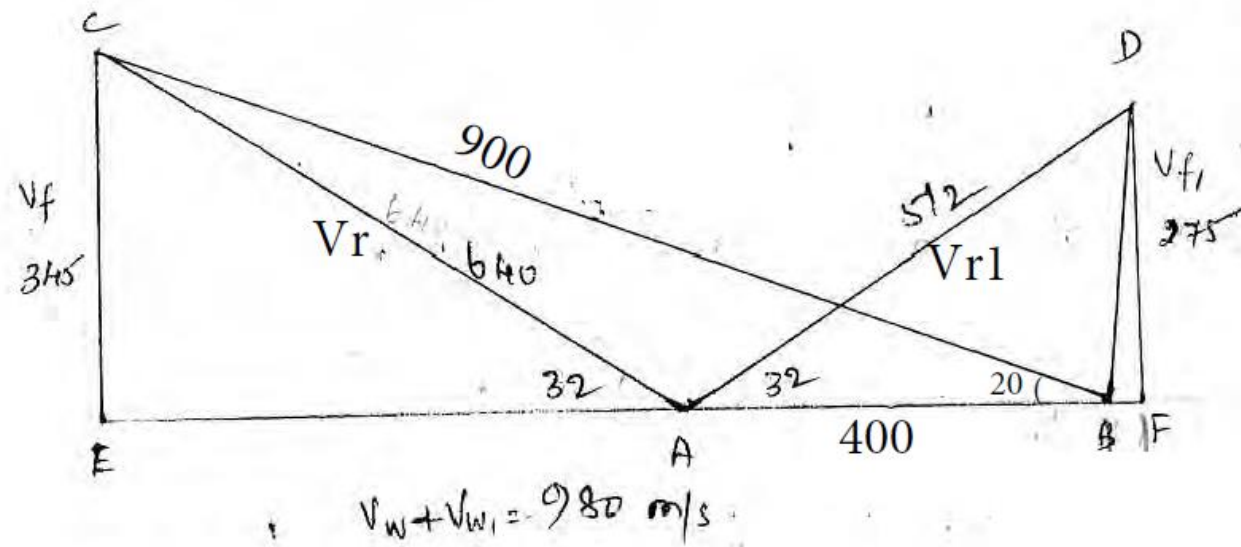
Since blades are symmetrical, blade angle α & ϕ are same.

Scale $1 \text{ cm} = 100 \text{ m/s}$.

$$V = 900 \text{ m/s} \quad \alpha = 20^\circ \quad V_b = 300 \text{ m/s}$$



$$V_w + V_{w1} = 980 \text{ m/s}$$

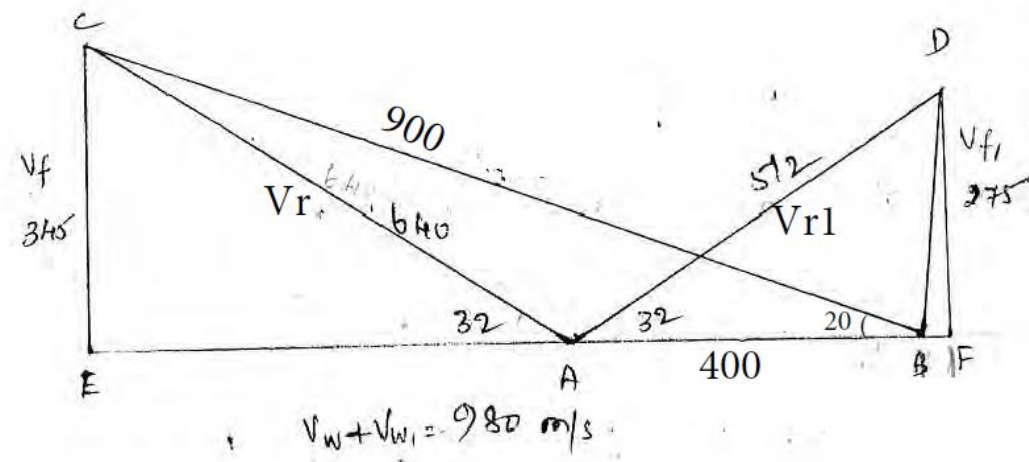


Blade angle at inlet $\theta = 32^\circ$

$$K = 0.8 = \frac{V_{r1}}{V_r}$$

$$V_{r1} = K \cdot V_r = 0.8 \times 640 = 512 \text{ m/s}$$

$$\text{Axial thrust} = m (V_f - V_{f1}) = \frac{1200}{3600} (345 - 275) = 23.33 \text{ N}$$



Power developed $P = \frac{m(V_w + V_{w1}) \cdot V_b}{1000} = \frac{1200}{3600} \times \frac{980}{1000} \times 400$

$= \underline{130.66 \text{ kW.}}$

Blade efficiency $\eta_b = \frac{2(V_w + V_{w1}) V_b}{V^2}$

$= \frac{2 \times 980 \times 400}{900^2} = 0.96 = 96.79 \%$